M6

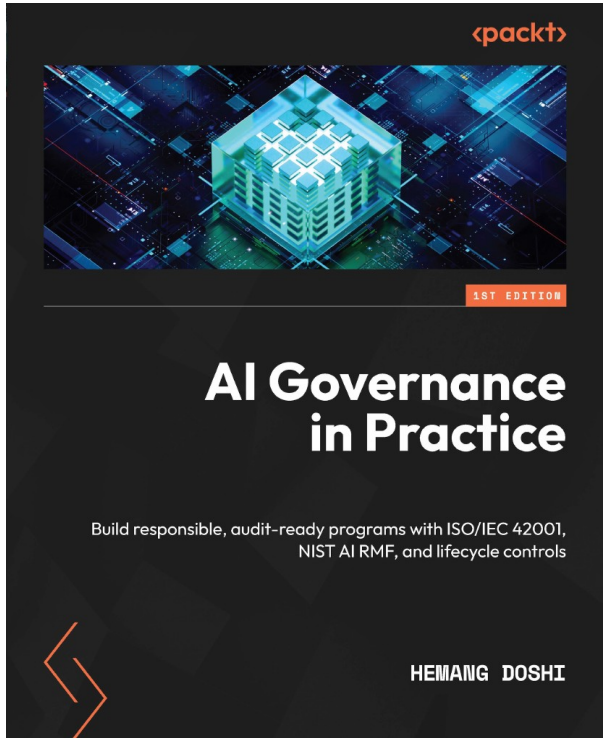
Foundations of AI Governance

Module 6 — From “how AI works” to “who is in charge”

307307 — BI Methods for Generative AI

AI Governance · University of Petra

Lecturer: _____ Date: _____



AI Governance in Practice — Hemang Doshi (Packt, 2026)

A Story to Start With

A fictional Monday morning

An email says your Citizen Credit Score has been lowered. You find out that you may lose unemployment support and bank loans, and that your insurance cost could triple. You call the office. The scoring rules cannot be shared, and there is no appeal.

1. **You cannot see**
   how the score is worked out

2. **You cannot understand**
   why it dropped

3. **You cannot challenge**
   the result or ask a person to review it

Question for the class: what would a well-governed version of this system need to have?

What You Will Be Able to Do

1. **Explain** how AI systems differ from normal software, and why that changes how we control them.
2. **Name** the three building blocks of an AI system and the main ways machines learn.
3. **Describe** what governance is, its key elements, and who does what.
4. **List** the twelve Responsible AI principles and give an example of one in practice.

AI Versus Normal Software

Aspect	Normal software	AI-based system
Decision logic	Fixed rules written by developers	Patterns learned from data
Learning	None	Learns and improves with more data
Over time	Behaves the same way	Behavior can change
New situations	Needs a code change	Can adapt from training data
Transparency	Logic is usually easy to follow	Logic can be complex or opaque
Data	Limited use of data	Depends heavily on large datasets

Based on Table 1.1 of the book.

Example: a fixed spam rule always flags the word “Free”. An AI filter learns when users move mail back to the inbox.

Why this matters

Normal software: review the code once, and it behaves the same until the code changes.

AI: the code can stay the same while the outputs change. Outputs must be watched all the time for errors, bias and drift.

Data, Algorithm, Model

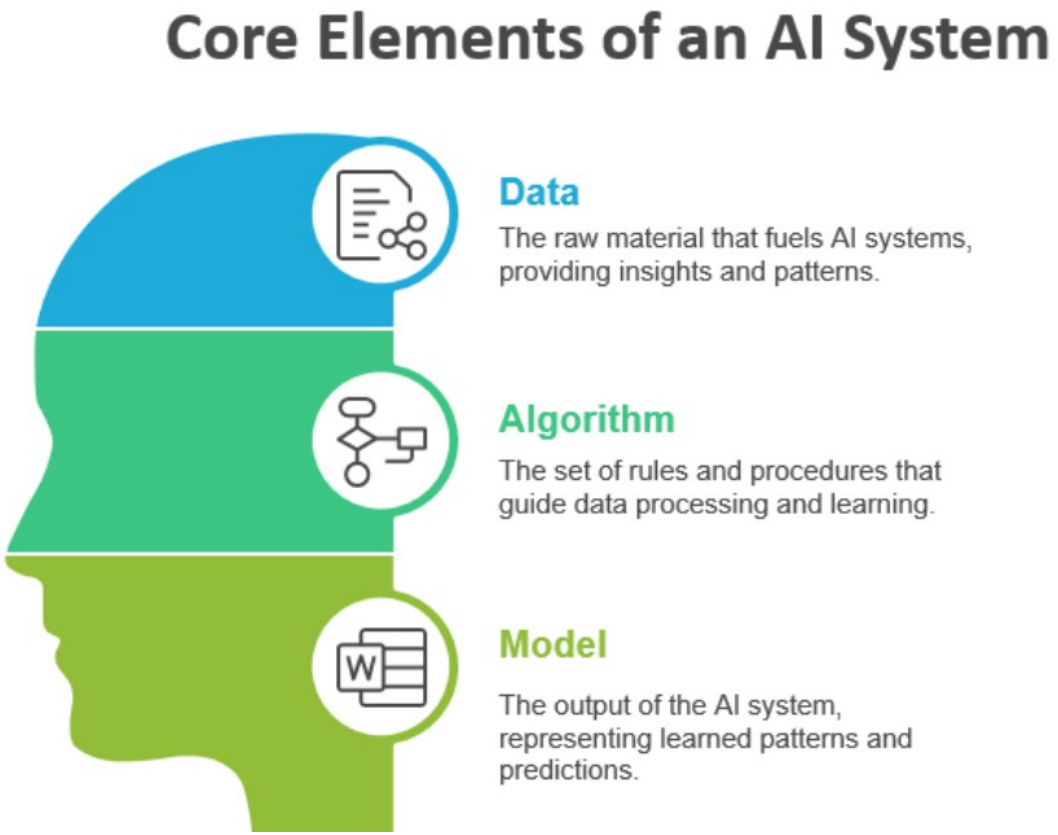


Figure 1.1 from AI Governance in Practice (Hemang Doshi, Packt, 2026).

Where risks begin

A model is only as reliable as the data and algorithm that built it.

- Data: gaps, bias, out-of-date records
- Algorithm: poor design or settings
- Model: misleading results and unfair decisions

Problems often start long before the model is finished.

How Machines Learn

1 Supervised

Learns from examples that already have the right answer (labels). Example: emails marked spam or not spam.

2 Unsupervised

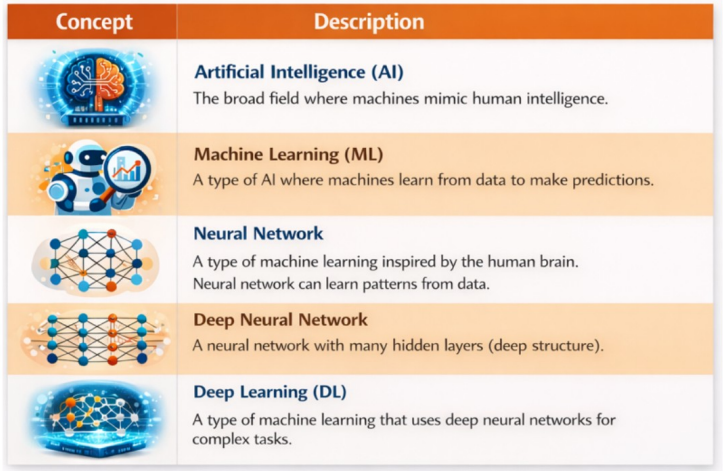
Finds hidden groups in data with no labels. Example: grouping customers by buying behavior.

3 Semi-supervised

A few labeled examples plus many unlabeled ones. Useful when labeling is costly.

4 Reinforcement

Learns by trial and error with rewards and penalties. Example: a robot learning a route.



Concept	Description
	Artificial Intelligence (AI) The broad field where machines mimic human intelligence.
	Machine Learning (ML) A type of AI where machines learn from data to make predictions.
	Neural Network A type of machine learning inspired by the human brain. Neural network can learn patterns from data.
	Deep Neural Network A neural network with many hidden layers (deep structure).
	Deep Learning (DL) A type of machine learning that uses deep neural networks for complex tasks.

Figure 1.3 from AI Governance in Practice (Hemang Doshi, Packt, 2026).

AI is the goal; machine learning is one way to get there

- Machine learning: machines learn from data instead of being told every rule
- Neural networks: layers of connected “neurons” that find patterns in images, speech and text
- Deep learning: neural networks with many layers, for complex tasks

Traditional ML and LLMs: Different Risks

Traditional ML models

Built for one purpose, such as credit scoring or fraud detection.

Governance focuses on: data quality, model validation, bias, explainability, performance, security and model changes.

Large language models (LLMs)

Broad capabilities, used for many purposes.

Extra risks: hallucinations, prompt attacks, data leakage, inappropriate outputs, misuse, copyright issues and third-party dependencies.

Where it runs matters too: cloud (easy to scale, but data sits with someone else), on-premises (more control, more effort) or edge (on the device itself: fast and works offline).

You met hallucinations and tool use in the agentic half. Here they become governance topics.

What Governance Is: Seven Elements

1 Governance committee

A cross-functional group that guides and oversees key decisions

2 Roles and responsibilities

Who owns decisions, outcomes and risks, written down

3 Policies and procedures

The rules, and how to follow them in practice

4 Decision-making

How use cases are approved, escalated or rejected

5 Risk management

Find, assess and manage legal, ethical and operational risks

6 Compliance management

Follow laws, regulations, standards and internal rules

7 Audit

An independent check that the controls really work

Governance =

direction, control and accountability

Layers of Responsibility and Rules

Governance

Sets the rules and boundaries

The committee decides that high-risk decisions need human review.

Management

Plans and organizes within them

Writes the plan, assigns teams, sets deadlines.

Operations

Does the day-to-day work

Configures the system, monitors it and sends high-risk cases to reviewers.

Policy

What we expect, and why

Procedure

Step-by-step how to do it

Guideline

Recommended practice, with flexibility

Laws and regulations come from outside. Governance turns them into the policies, procedures, roles and controls used inside the organization.

AI Governance Versus IT Governance

Aspect	IT governance	AI governance
Behavior	Stable and predictable once tested	Can change with retraining, drift or a new environment
Focus	Availability, security, cost, business fit	Adds ethics, bias, transparency, explainability, accountability
Reach	Mainly the organization	Also individuals, customers and society
Extra tools	Change management, security controls	AI ethics policy, impact assessments, human oversight, model monitoring

AI governance builds on IT governance; it does not replace it.

Why AI needs stronger governance

1. It learns and its behavior can change
2. Its decisions can be hard to explain, even for its developers
3. The rules about AI are changing fast

Who Does What

Board and senior management

Approve the framework and set the risk appetite

AI governance committee

Review and approve use cases, especially high-risk ones

Business and use-case owners

Define the purpose and own the impact of decisions

Data science and technical teams

Build, test, document and monitor the systems

Risk management

Identify, assess and report AI risks

Legal and compliance

Keep systems within laws, regulations and contracts

Information security and privacy

Protect data, models and access

Internal audit

Give independent assurance that governance works

Roles follow Chapter 2 of the book.

Responsible AI: Intelligence Is Not Enough



Figure 3.1 from AI Governance in Practice (Hemang Doshi, Packt, 2026).

Who defines the principles? Laws, standards bodies and industry groups. The board adopts them, and governance puts them to work.

1. Accountability
2. Fairness and non-discrimination
3. Transparency
4. Explainability
5. Privacy and data protection
6. Security and resilience
7. Safety and reliability
8. Human oversight
9. Risk-based approach
10. Ethical and responsible use
11. Continuous monitoring
12. Compliance with laws and standards

Case: Who Is Accountable?

A Company A

Builds a large language model (a general-purpose AI provider)

Documents the model's limits and known risks. Not directly responsible for loan decisions, unless hidden defects are proven.

B Company B

Uses that model to build a loan-approval system (the provider)

Must meet the rules for high-risk systems: risk management, data governance, testing, documentation. May share liability.

C Company C

A bank that uses the system to approve or reject loans (the deployer)

Primary accountability: human oversight, lawful use, monitoring outcomes, handling complaints, protecting rights.

Lesson: agree who is accountable before launch, through contracts, policies and risk assessments. Responsibility cannot be passed to the technology.

Case and likely outcome adapted from Chapter 3 of the book (illustrative, not legal advice).

Measuring Governance: KPIs

KPI area	What we measure	Target
AI inventory	Share of AI systems registered in the inventory	≥ 95%
Oversight	Share of AI use cases reviewed and approved by the committee	100%
Risk management	High-risk systems with a completed risk or impact assessment	100%
Review	System reviews completed on time	≥ 90%
Model oversight	Models monitored for bias and drift	≥ 95%
Incidents	Critical AI incidents reported	0
Issues	Average time to close governance issues	≤ 30 days
Training	Employees trained in governance and responsible AI use	≥ 90%

Example KPIs and thresholds based on Table 2.1 of the book. Each organization sets its own.

> *You cannot improve what you do not measure.*

Key Takeaways

1. AI learns from data and can change over time, so it cannot be governed like fixed-rule software.
2. Risks often begin early, in the data and the algorithm, not only in the finished model.
3. Governance is direction, control and accountability: committee, roles, policies, decisions, risk, compliance and audit.
4. Governance sets the rules, management plans, operations do the work; policy, procedure and guideline add the detail.
5. Responsible AI principles guide the choices. Accountability and human oversight must be clear before launch.

Discussion and Activity

Discuss

1. Which three principles matter most for a university chatbot, and why?
2. In the credit-score story, which governance element failed first?
3. Who should be accountable if the chatbot gives a student the wrong exam date?

Activity

Set up governance for Petra Study Buddy (a fictional campus AI assistant)

A. Is it AI?

B. Data, algorithm and model

C. Who does what?

D. Principles check

E. A mini policy and one KPI

Groups of 3–4. Handout: Module6_Foundations_activity.docx